Clustering Algorithms

Dr. Sameera Viswakula
sam@stat.cmb.ac.lk

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Clustering

Definition: Clustering is the process of grouping several objects into a number of groups, or clusters.

Goal: Objects in the same cluster are more similar to one another than they are to objects in other clusters.

Clustering - Example

Simple example

Species	Fat (%)	Proteins (%)
Horse	1.0	2.6
Donkey	1.4	1.7
Mule	1.8	2.0
Camel	3.4	3.5
Llama	3.2	3.9
Zebra	4.8	3.0
Sheep	6.4	5.6
Buffalo	7.9	5.9
Fox	5.9	7.4
Pig	5.1	6.6
Rabbit	13.1	7.1
Rat	12.6	12.3
Deer	19.7	9.2
Reindeer	20.3	10.4
Whale	21.2	11.1

Data taken from: W.S. Spector, Ed., Handbook of Biological Data, WB Saunders Company, London, 1956

Figure 1: Example

Clustering - Example..

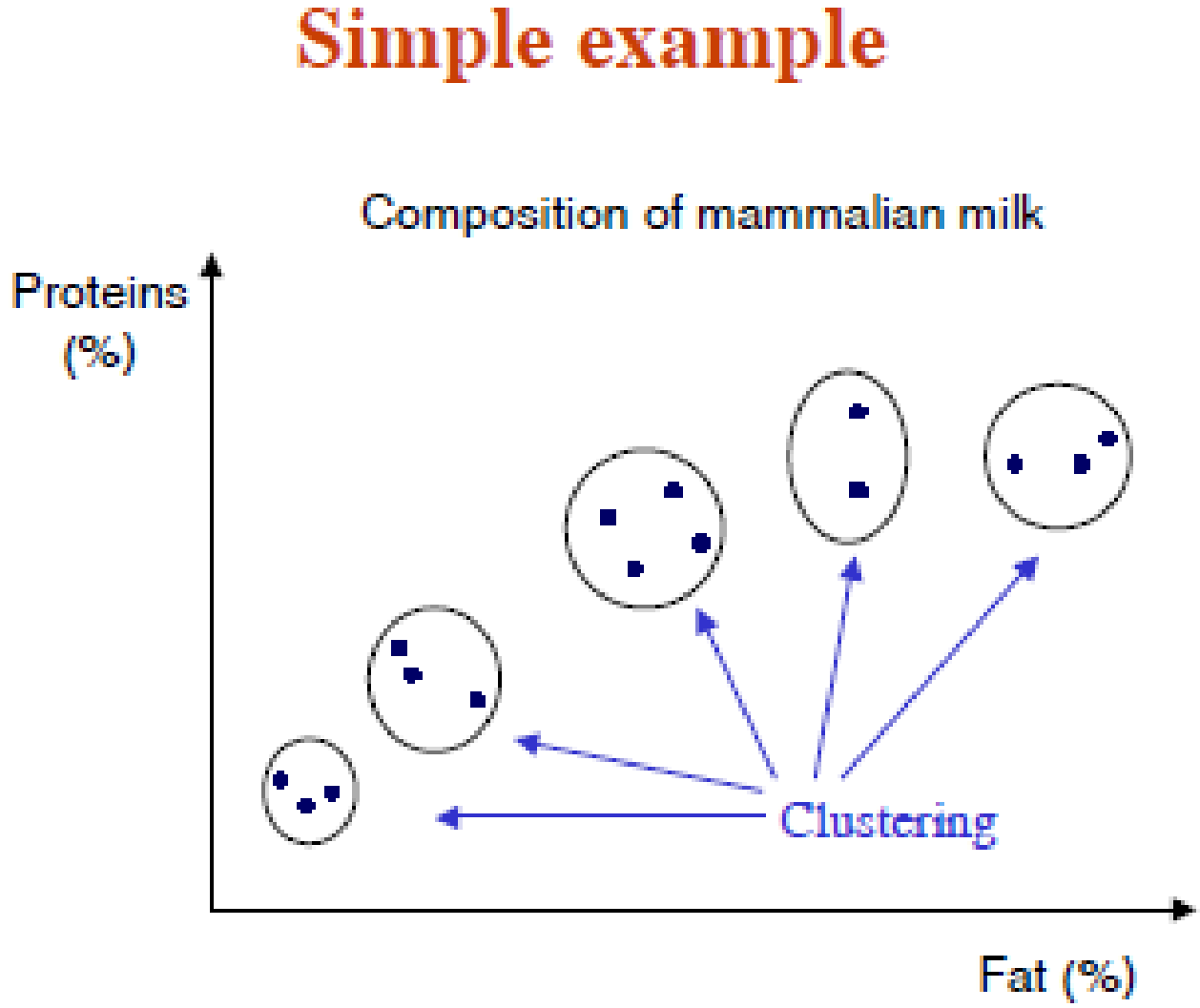


Figure 2: Identified clusters

Popular clustering algorithms

Some popular clustering algorithms are:

- K-means
- Hierarchical clustering
- Self organizing maps (SOM)
- Fuzzy clustering

Distance

The concept of distance plays a crucial role in all types of cluster analysis. For real numbers a and b a distance function d is defined as the absolute value of their difference

$$d(a, b) = |a - b| = \sqrt{(a - b)^2}$$

Note that

$$d(a, b) = |a - b| = \sqrt{(a - b)^2} = |b - a| = d(b, a)$$

Example 1: Let $a = 1$ and $b = 3$. Then $d(1, 3) = 2$.

Distance..

For the situation where gene expression values for several patients are available, it is of importance to define a distance for vectors of gene expressions such as $\mathbf{a} = (\mathbf{a_1}, ..., \mathbf{a_2})$ and $b = (b_1, ..., b_n)$. We shall concentrate mainly on the Euclidean distance, which is defined as the root of the sum of the squared differences

$$d(a, b) = \sqrt{\sum_{i=1}^{n} (a_i - b_i)^2}$$

.

Distance..

Example 2: Suppose that $\mathbf{a} = (\mathbf{a_1}, \mathbf{a_2}) = (\mathbf{1}, \mathbf{1})$ and $(\mathbf{b}) = (\mathbf{b_1}, \mathbf{b_2}) = (\mathbf{4}, \mathbf{5})$

$$d(a, b) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

$$= \sqrt{(1 - 4)^2 + (1 - 5^2)} = \sqrt{9 + 16} = 5.$$

Example 3: Compute the Euclidean distance between the following two vectors.

```
a <- c(1,1)
b <- c(4,5)
sqrt(sum((a-b)^2))
```

```
## [1] 5
```

Hierarchical clustering

Hierarchical clustering can be subdivided into two types:

- **Agglomerative clustering:** Here each observation is initially considered as a cluster of its own (leaf). Then, the most similar clusters are successively merged until there is just one single big cluster (root). i.e. bottom-up approach
- **Divise clustering:** This is begins with the root,i.e. all objects are included in one cluster. Then the most heterogeneous clusters are successively divided until all observation are in their own cluster. This is a top-down approach.

Our focus is on the agglomerative clustering method in this module. The result of hierarchical clustering is a tree-based representation of the objects, which is also known as dendrogram.

Steps to agglomerative hierarchical clustering

1. Preparing the data
2. Computing distance or (dis)similarity information between every pair of objects in the data set.
3. Using linkage function to group objects into hierarchical cluster tree, based on the distance information. Objects/clusters that are in close proximity are linked together using the linkage function.
4. Determining where to cut the hierarchical tree into clusters.

Merging mechanism

How to measure the similarity of two clusters? The linkage criterion determines the metric used for the cluster merging strategy:

- **Maximum or complete linkage** - minimizes the maximum distance between observations of pairs of clusters. Based on the similarity of the two least similar members of the clusters, this clustering tends to give tight spherical clusters as a final result.
- **Average linkage** - averages similarity between members, i.e., minimizes the average of the distances between all observations of pairs of clusters.
- **Ward linkage** - minimizes the sum of squared differences within all clusters. It is thus a variance-minimizing approach and in this sense is similar to the K-means objective function, but tackled with an agglomerative hierarchical approach.

Single Linkage Method

A cluster I is simply a set of data points $I = x_i$, where x_i is the i-th vector with gene expressions. In single linkage cluster analysis the distance between clusters I and J is defined as the smallest distance over all pairs of points of the two clusters:

$$d(I, J) = min_{i,j} d(x_i, x_j) : x_j \text{ in I and } x_j \text{ in } J.$$

Hence, the distance between the two clusters is the same as that of the nearest neighbors.

Single Linkage

The algorithm of single linkage cluster analysis starts with creating as many clusters as data points. Next, the nearest two are determined and these are merged into one cluster. Then the next two nearest clusters are determined and merged into one cluster. This process continuous until all points belong to one cluster.

Example 1:

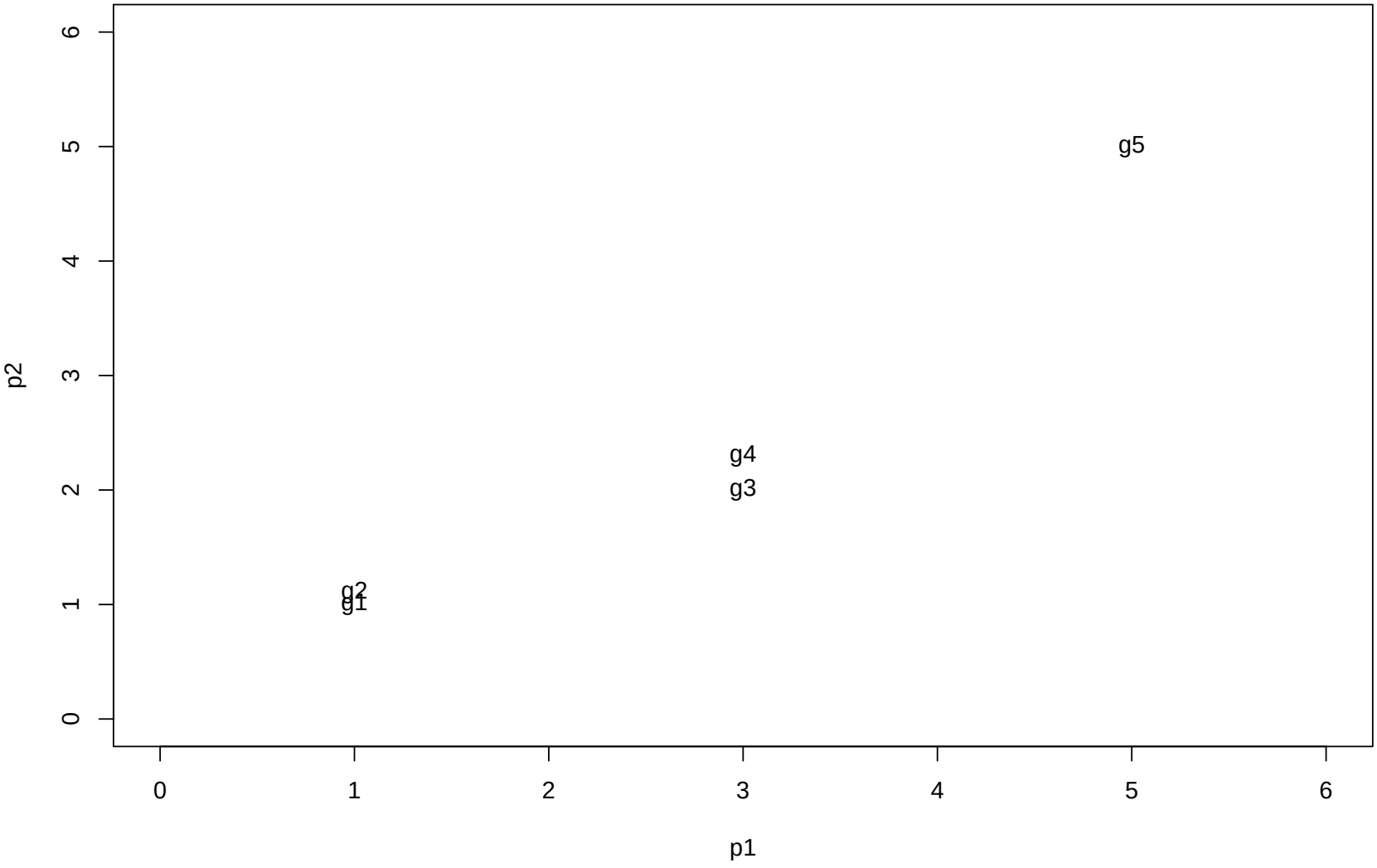
An explanatory example. To illustrate single linkage cluster analysis suppose we the following five gene expressions $g_1 = (1, 1)$, $g_2 = (1, 1.1)$, $g_3 = (3, 2)$, $g_4 = (3, 2.3)$, and $g_5 = (5, 5)$, from two persons. The expressions for each gene can be seen as coordinates on two perpendicular axis p_1 and p_2.

Single Linkage..

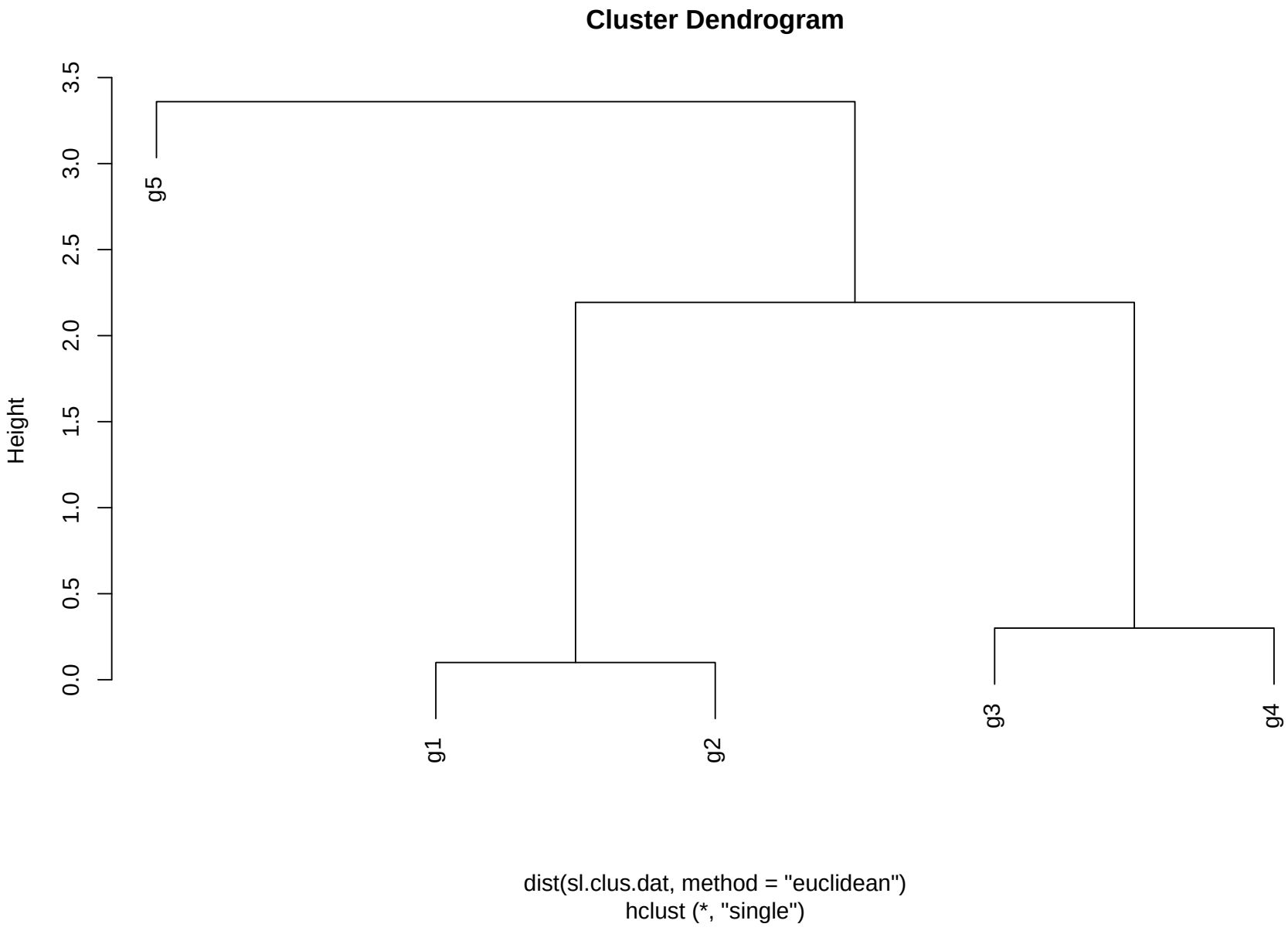
```
names <- list(c("g1","g2","g3","g4","g5"),c("p1","p2"))
sl.clus.dat <- matrix(c(1,1,1,1.1,3,2,3,2.3,5,5),ncol = 2,
byrow = TRUE,dimnames = names)
print(dist(sl.clus.dat,method="euclidean"),digits=3)
plot(sl.clus.dat,type="n", xlim=c(0,6), ylim=c(0,6))
text(sl.clus.dat,labels=row.names(sl.clus.dat))
```

```
     g1   g2   g3   g4
g2 0.10
g3 2.24 2.19
g4 2.39 2.33 0.30
g5 5.66 5.59 3.61 3.36
```

Plot of five points to be clustered



Tree of single linkage cluster analysis



Single Linkage - How it works

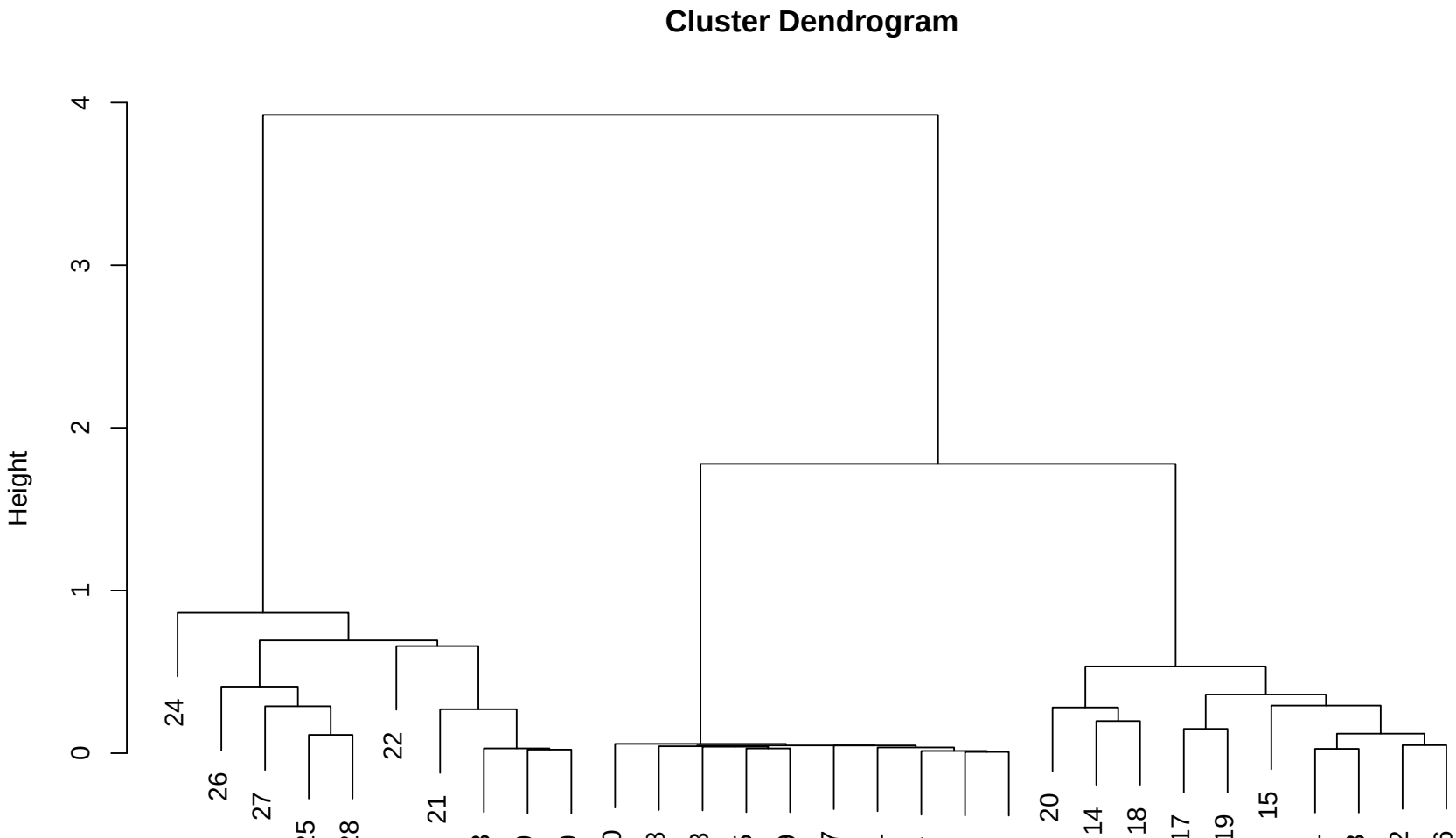
At the start each data point is seen as a separate cluster. Then the nearest two points (genes) from the Euclidian distance matrix are g_1 and g_2, having $d(g_1, g_2) = 0.10$. These two data points are merged into one cluster, say $I = \{g_1, g_2\}$. In previous figure this is illustrated by the horizontal line at height 0.10 in the tree. The other three data points g_3, g_4, g_5 are seen as three different clusters.

Next, the minimal distance between clusters can be read from the Euclidian distance matrix. Since the smallest is $d(g_3, g_4) = 0.30$, the new cluster $J = \{g_3, g_4\}$, corresponding to the horizontal line at height 0.30. Now there are three clusters, I, J, and $K = \{g_5\}$.

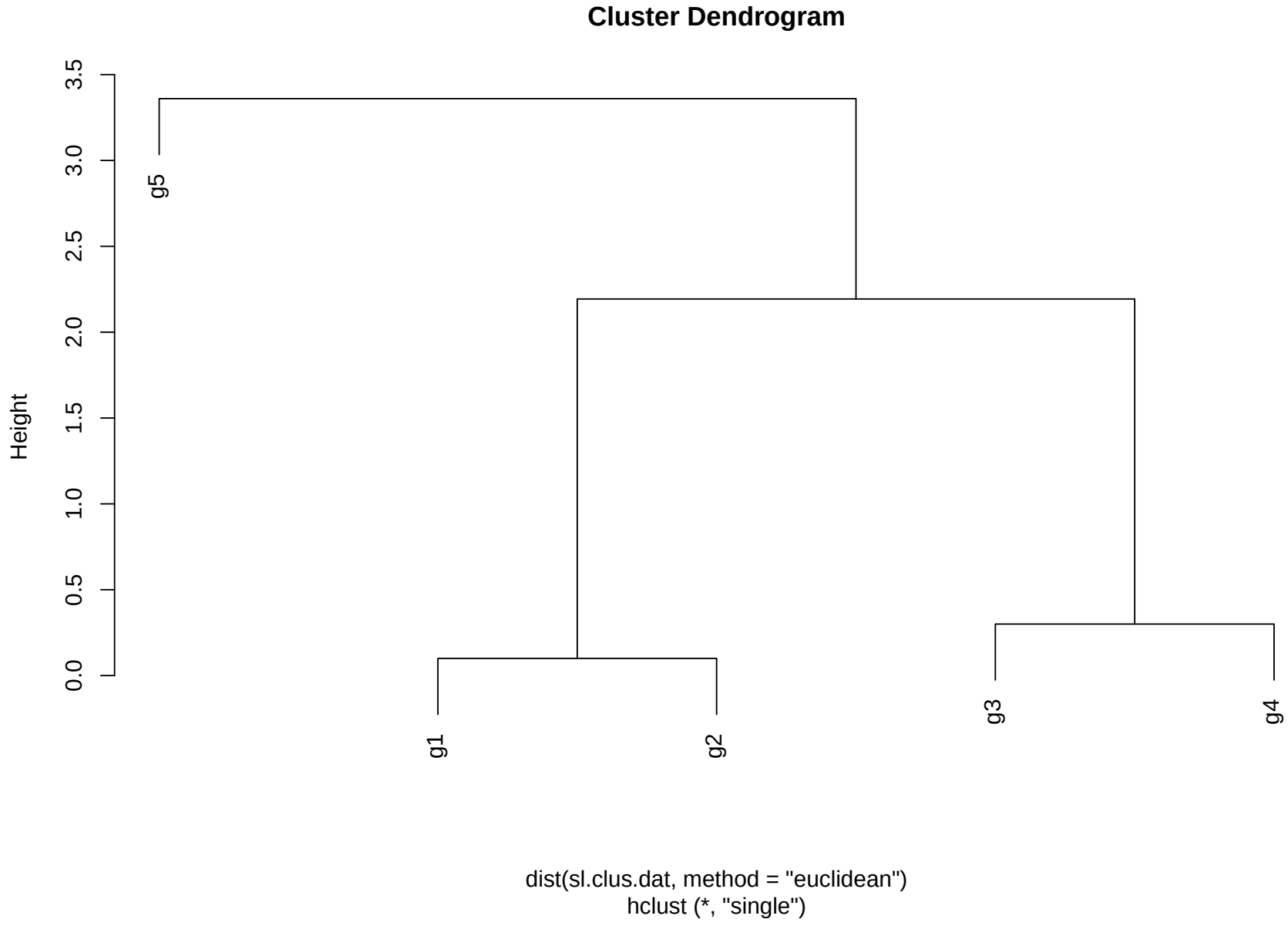
Hence, the cluster I and J are merged into one. Finally, the distance between cluster $\{g_1, g_2, g_3, g_4\}$, and the data point g_5 equals $d(g_4, g_5) = 3.36$, see the corresponding horizontal line at this height.

Example 2- Relating data generation processes to cluster trees.

```
x <- c(rnorm(10,0,0.1),rnorm(10,3,0.5),rnorm(10,10,1.0))
plot(hclust(dist(x,method="euclidean"),method="single"))
plot(sl.out)
```



Example 2- Relating data generation processes to cluster trees.



k-means Clustering

k-means cluster analysis is a popular method in bioinfomatics. It is defined by minimizing the within cluster sum of squares over K clusters. That is, given the data points $x_1, x_2, ..., x_n$ the method seeks to minimize the function

$$\sum_{k=1}^{K} \sum_{i \in I_k} d^2(x_i, a_k)$$

over all possible points $a_1, a_2, ..., a_k$.

This is accomplished by an algorithm (Hartigan & Wong, 1979) which starts by partitioning the data points into K initial clusters,eikther at random or using some heuristic device.

k-means Clustering

It then computes the cluster means (step 1) and constructs a new partition by associating each point with the closest cluster mean (step 2). The latter yields new clusters of which the means are calculated (step 1). Then it constructs a new partition by associating each point with the closest cluster mean (step 2). These two steps are repeated until convergence. The latter occurs when the data points no longer change clusters.

Example 3

Relating a data generation process to k-means cluster analysis. To illustrate k-means cluster analysis we shall simulate gene expressions from two different normal populations. That is, we randomly take fifty gene expressions for two persons from the $N(0, 0.5)$ population and fifty expressions for two persons from the $N(2, 0.5)$ population. The data points are collected in two matrices of order fifty by two which are placed one above the other. On the total of one hundred data points a $(k =)$ 2-means cluster analysis is performed.

Example 3..

```
data <- rbind(matrix(rnorm(50,0,0.5), ncol = 2),
              matrix(rnorm(50,2,0.5), ncol = 2))
cl <- kmeans(data, 2)
cl
```

Example 3..(part of the output)

```
K-means clustering with 2 clusters of sizes 25, 25

Cluster means:
        [,1]        [,2]
1 0.05041475 -0.01285769
2 1.91004858  2.09680534

Clustering vector:
 [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2
[39] 2 2 2 2 2 2 2 2 2 2 2 2

Within cluster sum of squares by cluster:
[1] 14.152367  8.678294
 (between_SS / total_SS =  81.2 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"
```

Example 3..

```
plot(data, col = cl$cluster)
points(cl$centers, col = 1:2, pch = 8, cex=2)
```

Ex: Obtain the dendrogram for the above dataset and identify the number of clusters.